Today

- The hydrogen atom
- General angular momentum
- Spin

4.4 The hydrogen atom

Question 4.4.1: What is the largest set of simultaneously diagonalizable operators among the position, linear momentum, and angular momentum operators in addition to the Hamiltonian for a system with central potential? (A). $\hat{H}$, $\hat{\vec{r}}$, $\hat{\vec{p}}$, $\hat{\vec{L}}$ (B). $\hat{H}$, $\hat{\vec{r}}$, $\hat{\vec{L}}$ (C). $\hat{H}$, $\hat{\vec{p}}$, $\hat{\vec{L}}$ (D). $\hat{H}$, $\hat{\vec{L}}^2$, $\hat{\vec{L}}_z$ (E). $\hat{H}$, $\hat{\vec{L}}^2$, $\hat{\vec{L}}_x$.

Exercise 4.4.1: Show that

1. $[\hat{\vec{L}}_z, \hat{\vec{p}}^2] = 0$,
2. $[\hat{\vec{L}}_z, \hat{V}(r)] = 0$

Answer: We could see that $[\hat{\vec{L}}_z, \hat{\vec{p}}^2] = 0$ without doing the math, since $\hat{\vec{p}}^2$ represents the magnitude (or length) of linear momentum squared, which is invariant under rotation. Moreover, $[\hat{\vec{L}}_z, \hat{V}(r)] = 0$, since a central potential involves only the length of the position vector, $r = |\vec{r}|$, which is invariant under rotation. Alternatively, we could do the math as follows.

1.

$$\begin{aligned}[\hat{\vec{L}}_z, \hat{\vec{p}}^2] &= [\hat{x}\hat{p}_y - \hat{y}\hat{p}_x, \hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2] = [\hat{x}, \hat{p}_x^2]\hat{p}_y - [\hat{y}, \hat{p}_y^2]\hat{p}_x \\ &= 2i\hbar\hat{p}_x\hat{p}_y - 2i\hbar\hat{p}_y\hat{p}_x = 2i\hbar[\hat{p}_x, \hat{p}_y] = 0. \end{aligned} \tag{4.4.1}$$

2.

$$[\hat{\vec{L}}_z, \hat{V}(r)] = [-i\hbar\frac{\partial}{\partial\phi}, \hat{V}(r)] = 0, \tag{4.4.2}$$

since ϕ and r independent spherical coordinates.

Question 4.4.2: Which of the following are true about a system with central potential? (A). Total angular momentum is conserved. (B). Each component of

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angular momentum is conserved. (C). Total linear momentum is conserved. (D). Each component of linear momentum is conserved.

For the specific case of the hydrogen atom, which contains one proton in the nucleus and one electron at distance r from it, the potential in position representation is

$$\hat{V} = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{r}. \tag{4.4.3}$$

The radial part of the Schrödinger equation given by (4.3.21)

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right)+\left(-\frac{1}{4\pi\epsilon_0}\frac{e^2}{r}+\frac{l(l+1)\hbar^2}{2mr^2}\right)R(r) = ER(r). \tag{4.4.4}$$

Note that the radial equation is a Schrödinger like equation with effective potential

$$V_{\text{eff}}(r) = V(r) + \frac{l(l+1)\hbar^2}{2mr^2} \tag{4.4.5}$$

with $V(r) = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{r}$. The second term, $\frac{l(l+1)\hbar^2}{2mr^2}$ arises as a centrifugal term because of angular momentum for $l \neq 0$. How the effective potential comes about as a sum of the Coulomb potential and the cetrifugal term are shown in the figure below.

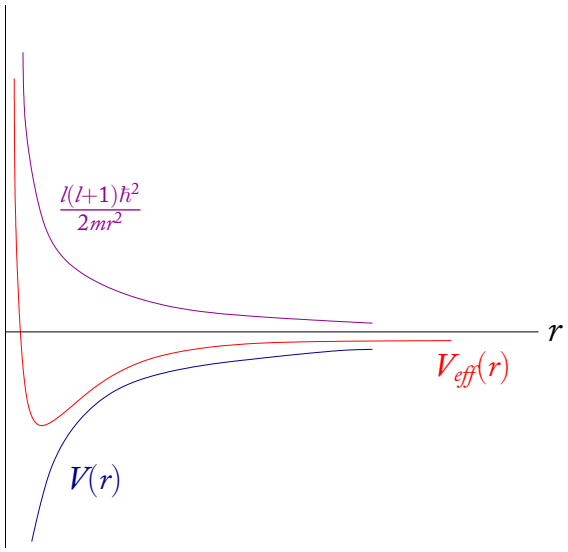


Figure 4.4.1: Effective potential for hydrogen atom.

Exercise 4.4.2:

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1. Show that the radial part of the Schödinger equation for the electron of mass m in a hydrogen atom with potential $V = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{r}$; i.e., $-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \left(-\frac{e^2}{4\pi\epsilon_0}\frac{1}{r} + \frac{\hbar^2}{2m}\frac{l(l+1)}{r^2}\right)u = Eu$, where $u(r) = rR(r)$, can be written as $\frac{d^2u}{d\rho^2} = \left(1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2}\right)u$, where $\rho = \kappa r$, $\rho_0 = \frac{me^2}{2\pi\epsilon_0\hbar^2\kappa}$, $\kappa = \sqrt{-\frac{2mE}{\hbar^2}}$, and l is the total angular quantum number.

2. Investigate the asymptotic large ρ behavior of the radial equation and show that normalizable solutions could be written as $u(\rho) = \rho^{l+1}e^{-\rho}v(\rho)$ with the equation $\rho\frac{d^2v}{d\rho^2} + 2(l+1-\rho)\frac{dv}{d\rho} + (\rho_0 - 2(l+1))v = 0$ for $v(\rho)$.

3. Show that the method of power series with $v(\rho) = \sum_{j=0}^{\infty} c_j\rho^j$ produces the recursion relations $c_{j+1} = \frac{2(j+l+1)-\rho_0}{(j+1)(j+2l+2)}c_j$.

4. Using the large j behavior of the recursion relation, argue how the infinite series would give $v(\rho) \sim e^{2\rho}$ and reintroduce a divergent solution.

5. Truncating the series, show that the energy levels are quantized and given by $E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2n^2} = -\frac{13.6\,eV}{n^2}$, where $n = 1, 2, 3, \cdots$. Show that the nth energy level is $2n^2-$fold degenerate.

6. Make a plot of the four lowest energy levels to scale, and indicate the list of associated quantum numbers (nlm_l) for each energy level.

7. Assuming that a hydrogen atom making a transition from a higher energy level with principal quantum number m to a lower energy level n gives off all its energy by releasing a photon, write down the expression for the frequency of the emitted radiation. Show that the possible frequencies can be written as $f = R_\infty\, c\,(\frac{1}{n^2} - \frac{1}{m^2})$, where $R_\infty \approx 1.1 \times 10^7 m^{-1}$ is the Rydberg constant and c is the speed of light.

The requirement of normalizable wavefunctions leads to quantized energy levels and the corresponding wavefunctions. As we see it in the problem set, the energies are given by

$$E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2n^2} = -\frac{13.6\,eV}{n^2}; \quad n = 1, 2, 3, \cdots \tag{4.4.6}$$

The energy levels can be rewritten in terms of the fine structure constant defined by

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \tag{4.4.7}$$

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and the Bohr radius given by

$$a = \frac{4\pi\epsilon_0\hbar^2}{me^2}, \tag{4.4.8}$$

as

$$E_n = -\frac{1}{2}\,\alpha^2\, mc^2 \frac{1}{n^2}. \tag{4.4.9}$$

or

$$E_n = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{2a}\frac{1}{n^2}.. \tag{4.4.10}$$

The wavefunctions are

$$\psi_{nlm_l}(r,\theta,\phi) = R_{nl}(r)Y_l^{m_l}(\theta,\phi), \tag{4.4.11}$$

with

$$R_{nl} = \frac{1}{r}\rho^{l+1}e^{-\rho}v(\rho), \tag{4.4.12}$$

where

$$v(\rho) = \sum_{j=0}^{j_{\max}} c_j\rho^j, \tag{4.4.13}$$

$$c_{j+1} = \frac{2(j+l+1)-\rho_0}{(j+1)(j+2l+2)}c_j, \tag{4.4.14}$$

and $j_{\max}$ is the order of the polynomial at which we truncate the series such that

$$2(j_{\max}+l+1)-\rho_0 = 0, \tag{4.4.15}$$

$$\rho_0 = \frac{me^2}{2\pi\epsilon_0\hbar^2}\sqrt{-\frac{\hbar^2}{2mE}} \tag{4.4.16}$$

$$j_{\max} = 0, 1, 2, ... \tag{4.4.17}$$

We define a new integer

$$n = j_{\max}+l+1 = \frac{\rho_0}{2}, \tag{4.4.18}$$

and

$$n = 1, 2, 3, ... \tag{4.4.19}$$

is called principal quantum number, which with (4.4.16) gives E_n. Furthermore, the angular momentum quantum number l for a fixed n takes possible values

$$l = 0, 1, 2, ..., n-1. \tag{4.4.20}$$

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The quantum number m_l for the $z-$component of angular momentum takes $2l+1$ possible values

$$m_l = -l, -m_l + 1, ..., l \tag{4.4.21}$$

The degeneracy of the nth energy level is

$$g(n) = \sum_{l=0}^{n-1} 2(2l+1) = 2n^2, \tag{4.4.22}$$

where the factor of 2 is because of two possible spin orientations (spin-up or spin-down) of the electron. The radial part of the wavefunction can be written as

$$v(\rho) = L_{n-l-1}^{2l+1}(2\rho), \tag{4.4.23}$$

where

$$L_{q-p}^{p}(x) \equiv (-1)^p \left(\frac{d}{dx}\right)^p L_q(x) \tag{4.4.24}$$

is called associated Laguerre polynomial with

$$L_q(x) \equiv e^x \frac{1}{q!}\left(\frac{d}{dx}\right)^q (e^{-x}x^q) \tag{4.4.25}$$

called the Laguerre polynomial.

The first few Laguerre polynomials are

$$\begin{aligned} L_0(x) &= 1 \\ L_1(x) &= -x + 1 \\ L_2(x) &= \frac{1}{2}x^2 - 2x + 1. \end{aligned} \tag{4.4.26}$$

The first few associated Laguerre polynomials are

$$\begin{aligned} L_0^0(x) &= 1 & L_0^2(x) &= 1 \\ L_1^0(x) &= -x + 1 & L_1^1(x) &= -x + 2 \\ L_0^1(x) &= 1 & L_1^2(x) &= -x + 3 \\ L_2^0(x) &= \frac{1}{2}x^2 - 2x + 1 & L_2^2(x) &= \frac{1}{2}x^2 - 4x + 6 \end{aligned} \tag{4.4.27}$$

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The first few radial wavefunctions are The first few Laguerre polynomials are

$$\begin{aligned} R_{10}(r) &= \frac{2}{\sqrt{a^3}} e^{-\frac{r}{a}} \\ R_{20}(r) &= \frac{1}{\sqrt{2a^3}} \left(1 - \frac{1}{2}\frac{r}{a}\right) e^{-\frac{r}{2a}} \\ R_{21}(r) &= \frac{1}{\sqrt{24a^3}} \frac{r}{a} e^{-\frac{r}{2a}}, \end{aligned} \tag{4.4.28}$$

where

$$a = \frac{4\pi\epsilon_0\hbar^2}{me^2} = 0.529 \times 10^{-10}\, m \tag{4.4.29}$$

is the Bohr radius.

The ground state wavefunction is

$$\psi_{100}(r,\theta,\phi) = \frac{1}{\sqrt{\pi a^3}} e^{-\frac{r}{a}} \tag{4.4.30}$$

Putting the radial and the angular parts together, the normalized wavefunctions of the hydrogen atom are

$$\psi_{nlm_l}(r,\theta,\phi) = \sqrt{\left(\frac{2}{na}\right)^3 \frac{(n-l-1)!}{2n(n+l)!}} e^{-\frac{r}{na}} \left(\frac{2r}{na}\right)^l \left[L^{2l+1}_{n-l-1}\left(\frac{2r}{na}\right)\right] Y_l^{m_l}(\theta,\phi). \tag{4.4.31}$$

These satisfy the orthonormality relation

$$\int \psi^*_{n'l'm'_l} \psi_{nlm_l} r^2 \sin\theta dr d\theta d\phi = \delta_{n'n}\delta_{l'l}\delta_{m'_l m_l}. \tag{4.4.32}$$

Demo 4.4.1: Hydrogen spectrum. We want to investigate the numerical values of radiation emitted in atomic transitions of hydrogen and find out which ones are in the visible range compared to what you see through the diffraction grating, and check if it matches what we see through a diffraction grating. We also view the spectra from He, Ne and Ar atoms in addition to that from an incandescent bulb.

Focusing on H atom, transitions that end in the ground state are called the Lyman series. Those that end in the first excited state are called the Balmer series. Those that end in the second excited state are called the Paschen series. These transitions are shown in the figure below.

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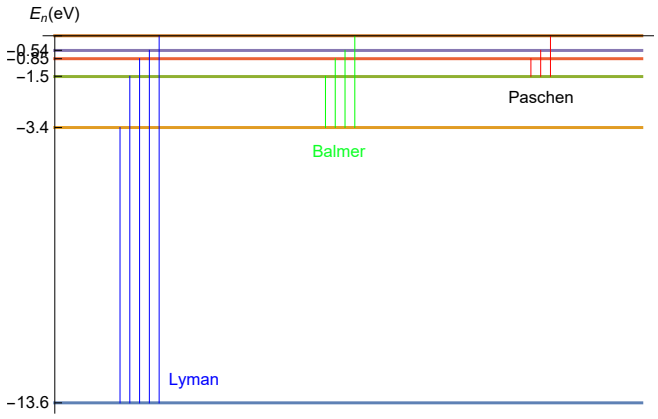


Figure 4.4.2: Hydrogen spectrum.

First, let us recall that red comes in $620 - 670\,nm$, blue in $450 - 495\,nm$, green in $495 - 570\,nm$, and violet in $380 - 450\,nm$. The transition from $m = 2$ to $n = 1$ gives $\Delta E = E_2 - E_1 = -3.4eV + 13.6eV = 10.2eV$, which corresponds to $\lambda = \frac{hc}{\Delta E} \approx 120nm$. This is the ultraviolet but not in the visible range. Because all other Lyman transitions have higher frequency and lower wavelength, none of them are in the visible range. Thus, we need to move to the Balmer series. The spectral lines we see using the diffraction grating correspond to individual transitions. We have diffraction gratings with 500 slits per millimeter. The angular separation for constructive interference patterns from a grating of separation d between slits is given

$$\sin\theta = \frac{\lambda}{d}, \tag{4.4.33}$$

where $d = \frac{1mm}{500} = 0.002mm$. We have the following table for the wavelengths in the Balmer series and the angles seen through the grating. We ignore the higher frequencies (shorter wavelengths), since they get too crowded together as the energy levels get very close, making it difficult to see them as separate spectral lines. These look like a single line.

ΔE	$\lambda = \frac{hc}{\Delta E}$	Color	Angle
$E_3 - E_2 = 1.89\,eV$	$656\,nm$	red	18.8^0
$E_4 - E_2 = 2.55\,eV$	$486\,nm$	cyan (blue-green)	13.9^0
$E_5 - E_2 = 2.86\,eV$	$434\,nm$	violet	12.4^0
$E_6 - E_2 = 3.02\,eV$	$410\,nm$	violet	11.8^0

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It is important to note the differences in the spectra from the incandescent bulb and H, He, Ne, and Ar atoms. The emission from the incandescent bulb is due to thermal radiation arising from the jiggling motion of charges, and it contains a broad spectrum of frequencies. On the other hand, the spectrum from the atoms arises from transitions between discrete energy levels, resulting in spectral lines. Note the red color in H atom has a specific location that is different from a red in another atom, such as He. This has applications in various fields such as observational astronomy, where the spectroscopy of radiation could be used to find out the atomic composition of the surface of a star.

Question 4.4.3: The probability density for finding the electron at the origin in the ground state of H atom is (A). $|\psi_{100}(0,\theta,\phi)|^2$, (B). $|R_{10}(0)|^2$, (C). 0.

Question 4.4.4: How many total number of nodes does the wavefunction $\psi_{nlm_l}(r,\theta,\phi)$ of H atom has? (A). n, (B). l, (C). m_l, (D). $n-1$.

4.5 General angular momentum

Now we want to investigate angular momentum based on the commutation relations and without consideration of representation either in position or momentum space. We call such angular momentum general angular momentum and denote it by $\hat{\vec{J}}$. In terms of its cartesian components

$$\hat{\vec{J}} = (\hat{J}_x, \hat{J}_y, \hat{J}_z) = (\hat{J}_1, \hat{J}_2, \hat{J}_3), \tag{4.5.1}$$

where we interchangeably use the indices $1,2,3$ for x,y,z. For the total angular momentum we write

$$\hat{\vec{J}}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2. \tag{4.5.2}$$

We take the commutation relations are

$$[\hat{J}_i, \hat{J}_j] = i\hbar\epsilon_{ijk}\hat{J}_k \tag{4.5.3}$$

as the definition of angular momentum. Moreover,

$$[\hat{\vec{J}}^2, \hat{J}_i] = 0. \tag{4.5.4}$$

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We take the commuting set to be $\hat{\vec{J}}^2$ and $\hat{J}_z$. Now suppose we trade $\hat{J}_x$ and $\hat{J}_y$ for

$$\hat{J}_\pm = \hat{J}_x \pm i\hat{J}_y. \tag{4.5.5}$$

The commutation relations become

$$\begin{aligned}[\hat{J}_+, \hat{J}_-] &= [\hat{J}_x + i\hat{J}_y, \hat{J}_x - i\hat{J}_y] \\ &= -i[\hat{J}_x, \hat{J}_y] + i[\hat{J}_y, \hat{J}_x] = -i(i\hbar\hat{J}_z) + i(-i\hbar\hat{J}_z) = 2\hbar\hat{J}_z.\end{aligned} \tag{4.5.6}$$

$$\begin{aligned}[\hat{J}_z, \hat{J}_\pm] &= [\hat{J}_z, \hat{J}_x \pm i\hat{J}_y] \\ &= [\hat{J}_z, \hat{J}_x] \pm i[\hat{J}_z, \hat{J}_y] = i\hbar\hat{J}_y \pm i(i\hbar\hat{J}_x) = \pm\hbar(\hat{J}_x \pm i\hat{J}_y) = \pm\hbar\hat{J}_\pm.\end{aligned} \tag{4.5.7}$$

Moreover, because, $\hat{\vec{J}}^2$ commutes with every components of $\hat{\vec{J}}$,

$$[\hat{\vec{J}}^2, \hat{J}_\pm] = 0. \tag{4.5.8}$$

Because $\hat{\vec{J}}^2$ and $\hat{J}_z$ commute, we can simultaneously diagonalize them. Let the simultaneous eigenstate be $|ab\rangle$ such that

$$\begin{aligned}\hat{\vec{J}}^2|ab\rangle &= a|ab\rangle \\ \hat{J}_z|ab\rangle &= b|ab\rangle\end{aligned} \tag{4.5.9}$$

Question 4.5.1: Which of the following statements are true? (A). $\hat{J}_+$ raises the eigenvalue a. (B). $\hat{J}_+$ raises the eigenvalue b. (C). $\hat{J}_-$ lowers the eigenvalue a. (D). $\hat{J}_-$ lowers the eigenvalue b.

First,

$$\hat{\vec{J}}^2\left(\hat{J}_\pm|ab\rangle\right) = \left([\hat{\vec{J}}^2, \hat{J}_\pm] + \hat{J}_\pm\hat{\vec{J}}^2\right)|ab\rangle = \left(0 + a\hat{J}_\pm\right)|ab\rangle = a\left(\hat{J}_\pm|ab\rangle\right). \tag{4.5.10}$$

Therefore, $\hat{J}_\pm|ab\rangle$ is an eigenstate of $\hat{\vec{J}}^2$ with the same eigenvalue a as $|ab\rangle$ and does not raise or lower a, as expected, since $[\hat{\vec{J}}^2, \hat{J}_\pm] = 0$. On the other hand,

$$\hat{J}_z\left(\hat{J}_\pm|ab\rangle\right) = \left([\hat{J}_z, \hat{J}_\pm] + \hat{J}_\pm\hat{J}_z\right)|ab\rangle = \left(\pm\hbar\hat{J}_z + b\hat{J}_\pm\right)|ab\rangle = (b \pm \hbar)\left(\hat{J}_\pm|ab\rangle\right). \tag{4.5.11}$$

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Thus, $\hat{J}_+$ raises and $\hat{J}_-$ lowers the eigenvalue a of $\hat{J}_z$. Therefore, assuming $|ab\rangle$ is normalized,

$$\hat{J}_\pm|ab\rangle = c_\pm|ab \pm \hbar\rangle, \tag{4.5.12}$$

where $c_\pm$ are normalization constants.

Note that

$$\begin{aligned}\hat{J}_\pm\hat{J}_\mp &= (\hat{J}_x \pm i\hat{J}_y)(\hat{J}_x \mp i\hat{J}_y)\\ &= \hat{J}_x^2 + \hat{J}_y^2 \mp i\hat{J}_x\hat{J}_y + \pm i\hat{J}_y\hat{J}_x = \hat{J}_x^2 + \hat{J}_y^2 \mp i[\hat{J}_x, \hat{J}_y]\\ &= \hat{J}_x^2 + \hat{J}_y^2 \pm \hbar\hat{J}_z = \hat{\vec{J}}^2 - \hat{J}_z^2 \pm \hbar\hat{J}_z,\end{aligned} \tag{4.5.13}$$

or

$$\hat{J}_\mp\hat{J}_\pm = \hat{J}_x^2 + \hat{J}_y^2 \mp \hbar\hat{J}_z = \hat{\vec{J}}^2 - \hat{J}_z^2 \mp \hbar\hat{J}_z, \tag{4.5.14}$$

and

$$\frac{1}{2}\left(\hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+\right) = \hat{J}_x^2 + \hat{J}_y^2 = \hat{\vec{J}}^2 - \hat{J}_z^2. \tag{4.5.15}$$

In particular,

$$\hat{J}_-\hat{J}_+ = \hat{\vec{J}}^2 - \hat{J}_z^2 - \hbar\hat{J}_z, \qquad \hat{J}_+\hat{J}_- = \hat{\vec{J}}^2 - \hat{J}_z^2 + \hbar\hat{J}_z. \tag{4.5.16}$$

Now consider states with state $|ab_{\max}\rangle$ having the highest value of $b = b_{\max}$ such that $\hat{J}_+|ab_{\max}\rangle = 0$ and $|ab_{\min}\rangle$ having the lowest value of $b = b_{\min}$ such that $\hat{J}_-|ab_{\min}\rangle = 0$.

And then, taking the inner product of $\hat{J}_+|ab_{\max}\rangle$ with itself,

$$\begin{aligned}0 &= \langle ab_{\max}|\hat{J}_-\hat{J}_+|ab_{\max}\rangle = \langle ab_{\max}|\hat{\vec{J}}^2 - \hat{J}_z^2 - \hbar\hat{J}_z|ab_{\max}\rangle\\ &= a - b_{\max}^2 - \hbar b_{\max}.\end{aligned} \tag{4.5.17}$$

In other words,

$$b_{\max} = \frac{-\hbar \pm \sqrt{\hbar^2 + 4a}}{2} \tag{4.5.18}$$

Furthermore, taking the inner product of $\hat{J}_-|ab_{\min}\rangle$ with itself,

$$0 = \langle ab_{\min}|\hat{J}_+\hat{J}_-|ab_{\min}\rangle = \langle ab_{\min}|\hat{\vec{J}}^2 - \hat{J}_z^2 + \hbar\hat{J}_z|ab_{\min}\rangle$$

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$$= \quad a - b_{\text{min}}^2 + \hbar b_{\text{min}}. \tag{4.5.19}$$

In other words,

$$b_{\text{min}} = \frac{\hbar \pm \sqrt{\hbar^2 + 4a}}{2} \tag{4.5.20}$$

Because we require $b_{\text{max}} \geq b_{\text{min}}$, we choose the plus sign in b_{max} and the minus sign in b_{min}; i.e.,

$$b_{\text{max}} = \frac{-\hbar + \sqrt{\hbar^2 + 4a}}{2}, \qquad b_{\text{min}} = \frac{\hbar - \sqrt{\hbar^2 + 4a^2}}{2}. \tag{4.5.21}$$

Note that $b_{\text{min}} = -b_{\text{max}}$ Because the value of b increases in steps of $\hbar$, we must have

$$b_{\text{max}} = b_{\text{min}} + n\hbar, \tag{4.5.22}$$

where n is a non-negative integer, $n = 0, 1, 2, \ldots$ Moreover, because of isotropy of space the maximum value for the eigenvalue of $\hat{L}_z$ in one frame could be converted to a minimum value in another, or just using (4.5.21),

$$b_{\text{min}} = -b_{\text{max}}. \tag{4.5.23}$$

Thus

$$b_{\text{max}} = -b_{\text{max}} + n\hbar \tag{4.5.24}$$

or

$$b_{\text{max}} = -b_{\text{min}} = \frac{n}{2}\hbar, \qquad n = 0, 1, 2, \ldots \tag{4.5.25}$$

We define $j = \frac{n}{2}$ and rewrite this as

$$b_{\text{max}} = -b_{\text{min}} = j\hbar, \qquad j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \ldots \tag{4.5.26}$$

The value of a then follows from

$$a = b_{\text{max}}^2 + \hbar b_{\text{max}} = j^2\hbar^2 + \hbar^2 j = j(j+1)\hbar^2. \tag{4.5.27}$$

Question 4.5.2: What important difference do you see between orbital and general angular momentum? Which of the following are correct regarding orbital and general angular momenta? (A). Total orbital angular momentum can take $j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$, (B). Total angular momentum can take $j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$, (C). Total orbital angular momentum can take only $j = 0, 1, 2, \ldots$, (A). Total angular momentum can take only $j = 0, 1, 2, \ldots$

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Thus, for a given $b_{\max} = j\hbar$ or $a = j(j+1)\hbar^2$, the possible values of b range from $b_{\min} = -j\hbar$ to $b_{\max} = j\hbar$ in steps of $\hbar$. In other words, b takes values $m_j\hbar$, where

$$m_j = -j, -j+1, ..., j. \tag{4.5.28}$$

To determine the normalization constants, let us first rewrite (4.5.12) as

$$\hat{J}_\pm |jm_j\rangle = c^\pm_{jm_j} |jm_j \pm 1\rangle, \tag{4.5.29}$$

The inner product is

$$\langle jm_j|\hat{J}_\mp \hat{J}_\pm|jm_j\rangle = |c^\pm_{jm_j}|^2 \langle jm_j \pm 1|\hat{J}_\mp \hat{J}_\pm|jm_j \pm 1\rangle = |c^\pm_{jm_j}|^2 \tag{4.5.30}$$

On the other hand,

$$\langle jm_j|\hat{J}_\mp \hat{J}_\pm|jm_j\rangle = \langle jm_j|\hat{\vec{J}}^2 - \hat{J}_z^2 \mp \hbar \hat{J}_z|jm_j\rangle = j(j+1)\hbar^2 - m_j^2\hbar^2 \mp \hbar^2 m_j. \tag{4.5.31}$$

Therefore,

$$|c^\pm_{jm_j}|^2 = j(j+1)\hbar^2 - m_j^2\hbar^2 \mp \hbar^2 m_j = \hbar^2(j \mp m_j)(j \pm m_j + 1) \tag{4.5.32}$$

Choosing real normalization constants,

$$\hat{J}_\pm |jm_j\rangle = \hbar \sqrt{(j \mp m_j)(j \pm m_j + 1)}\, |jm_j \pm 1\rangle. \tag{4.5.33}$$

Note that the matrix elements of $\hat{J}_\pm$ are

$$\begin{aligned} \langle j' m'_j|\hat{J}_\pm|jm_j\rangle &= \hbar \sqrt{(j \mp m_j)(j \pm m_j + 1)} \langle j' m'_j|jm_j \pm 1\rangle \\ &= \hbar \sqrt{(j \mp m_j)(j \pm m_j + 1)}\, \delta_{j'j}\delta_{m'_j, m_j \pm 1}. \end{aligned} \tag{4.5.34}$$

4.6 Spin

4.6.1 Classical orbital and spinning motion

The revolution of the earth around the sun is an example of a classical orbital motion. The rotation of the earth about an axis through it is an example of a spinning motion. In either case, the infinitesimal classical angular momentum corresponding to an infinitesimal linear momentum $d\vec{p} = dm\vec{v}$ of mass dm at position $\vec{r}$ from the axis of rotation moving with velocity $\vec{v}$ is given

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by

$$d\vec{L} = \vec{r} \times d\vec{p}. \tag{4.6.1}$$

The total angular momentum is

$$\vec{L} = \int \vec{r} \times d\vec{p} = \int \vec{r} \times dm\vec{v}. \tag{4.6.2}$$

In particular, for a point particle of mass m at fixed radius r spinning with constant angular velocity ω about the $z-$axis,

$$\vec{L} = I\omega\hat{z} = m\omega r^2\hat{z}. \tag{4.6.3}$$

If the rotating particle has charge q, it also has a magnetic dipole moment $\vec{\mu}$. Together with the angular momentum $\vec{L}$,

$$\vec{L} = m\omega r^2\hat{z}, \qquad \vec{\mu} = \frac{q}{2}\omega r^2\hat{z}, \tag{4.6.4}$$

where we used $\mu = I\pi r^2$ with current $I = \frac{q}{2\pi/\omega} = \frac{q\omega}{2\pi}$. Thus,

$$\vec{\mu} = \frac{q}{2m}\vec{L}. \tag{4.6.5}$$

Exercise 4.6.1:

1. Consider a uniformly charged wire ring of mass M and total charge Q rotating about its axis. Find the ratio of its magnetic dipole moment to its angular momentum. This is called the gyromagnetic ratio. (5 points)
2. Generalize your result in part (1) for the gyromagnetic ratio of a uniformly charged spinning sphere of mass M carrying charge Q. (5 points)
3. Supposing that the electron is a uniformly charged sphere and using your result in part (2), calculate the numerical value of its gyromagnetic ratio. (5 points)
4. Experiment puts an upper bound of about 10^{-22} m to the radius of the electron. Supposing that the electron is a spherical shell of uniform mass density and assuming that the farthest point on the surface spins at the speed of light, find an upper bound to the classical angular momentum of the electron. Using this and the gyromagnetic ratio in part (3), find the corresponding numerical value of the magnetic dipole moment of the electron. (5 points)

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5. Experiment shows that the electron has intrinsic angular momentum called spin with value of $\frac{\hbar}{2}$, where $\hbar = \frac{h}{2\pi}$ and h is Planck's constant, and that the gyromagnetic ratio is bigger by about a factor of 2 called the $g-$factor. The current experimental value of the electron spin $g-$factor is 2.00231930436256(35), and the experimental value of the magnetic dipole moment of the electron is $[-928.476377 \times 10^{-26} \pm (0.000023 \times 10^{-26})]JT^{-1}$. How does the classical angular momentum in part (4) compare to the experimental value of spin, and what does this tell us about the possibility of a classical interpretation to spin? (5 points)

Solution:

1. The magnitude of magnetic dipole moment is $\mu = Ia = \frac{Q}{T}\pi R^2 = \frac{Qv}{2\pi R}\pi R^2 = \frac{1}{2}QvR$ and the angular momentum is $L = MvR$. The gyromagnetic ratio is then $g = \frac{m}{L} = \frac{Q}{2M}$.
2. We can decompose the sphere into rings with infinitesimal thickness and, because g does not depend on the radius of the rings, the gyromagnetic ratio is $g = \frac{Q}{2M}$.
3. For the electron $g = \frac{-e}{2m_e} = -8.8 \times 10^{10} C.kg^{-1}$.
4. The angular momentum is $L = I\omega = \frac{2}{3}m_e r_e^2 \frac{c}{r_e} = \frac{2}{3}m_e r_e c \leq \frac{2}{3}(9.11 \times 10^{-31}kg)(10^{-22}m)(3 \times 10^{-8} m.s^{-1}) = 1.8 \times 10^{-44} J.s^{-1}$. The magnetic dipole moment is $\mu = gL \leq 1.6 \times 10^{-33} J.T^{-1}$.
5. The classical value of angular momentum is much smaller than the measured value, $\frac{L}{\hbar/2} \leq 3.5 \times 10^{-10}$. Thus the spin of the electron cannot be explained by a classical spinning motion. [Aside remark: Spin is one of the defining intrinsic properties of elementary particles, just like electric charge and mass.]

4.6.2 Spin

Spin is a quantum mechanics property with no classical analogue. In particular, as we see in Ex. 4.6.1, the magnitude of the intrinsic angular momentum of an electron is too large to be reproduced by a classical spinning motion. A measurement of the spin of an electron along a given direction, such as along z, could produce either one of only two possible values. Thus, the electron has $j = \frac{1}{2}$ and $m_j = \pm\frac{1}{2}$.

In other words, the special case of $j = \frac{1}{2}$ describes the spin (or spin angular momentum) of a

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particle of spin$-\frac{1}{2}$, and we often denote it by $s = \frac{1}{2}$. We also often denote the spin angular momentum operator by $\hat{\vec{S}}$ in place of $\hat{\vec{J}}$. The possible values of m_s are $\frac{1}{2}$ and $-\frac{1}{2}$. Let us denote the state with $s = \frac{1}{2}$ and $m_s = \frac{1}{2}$ (spin-up along z) by $|\frac{1}{2}, \frac{1}{2}\rangle$ and the state with $s = \frac{1}{2}$ and $m_s = -\frac{1}{2}$ (spin-down along z) by $|\frac{1}{2}, -\frac{1}{2}\rangle$, where we have started to put a comma between the quantum numbers j and m_j in $|j, m_j\rangle$ for clarity of notation, which we will often follow from now on. Because the spin has two possible components (up or down), the Hilbert space of a spin-half system is two-dimensional and complex. The basis vectors are

$$|\frac{1}{2}, \frac{1}{2}\rangle \equiv |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |\frac{1}{2}, -\frac{1}{2}\rangle \equiv |2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{4.6.6}$$

Let us construct the matrices for $\hat{S}_x$, $\hat{S}_y$, $\hat{S}_z$, and $\hat{\vec{S}}^2$. First, the matrices $\hat{S}_z$ and $\hat{\vec{S}}^2$ are diagonal, since they are the ones that are simultaneously diagonalized. The matrix elements are

$$\langle s' m_s' | \hat{\vec{S}}^2 | s m_s \rangle = s(s+1)\hbar^2 \delta_{s's} \delta_{m_s' m_s}, \qquad \langle s' m_s' | \hat{S}_z | s m_s \rangle = m_s \hbar \delta_{s's} \delta_{m_s' m_s} \tag{4.6.7}$$

In matrix form,

$$\hat{\vec{S}}^2 = \frac{3}{4}\hbar^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{4.6.8}$$

Using the matrix elements for $\hat{S}_\pm$, we have

$$\hat{S}_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \hat{\vec{S}}^2 = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \tag{4.6.9}$$

Then, using $\hat{S}_x = \frac{1}{2}(\hat{S}_+ + \hat{S}_-)$ and $\hat{S}_x = \frac{1}{2i}(\hat{S}_+ - \hat{S}_-)$, we have

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{4.6.10}$$

The spin angular momentum can be rewritten as

$$\vec{S} = \frac{\hbar}{2}\vec{\sigma}, \tag{4.6.11}$$

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where the components of $\vec{\sigma}$ are called the Pauli matrices, and given by

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{4.6.12}$$

Observe that

$$[\hat{S}_i, \hat{S}_j] = i\hbar\epsilon_{ijk}\hat{S}_k \tag{4.6.13}$$

and

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k. \tag{4.6.14}$$

A general state of a spin-half system is then given by

$$|\chi\rangle = a|\frac{1}{2}, \frac{1}{2}\rangle + b|\frac{1}{2}, \frac{1}{2}\rangle = \begin{pmatrix} a \\ b \end{pmatrix}, \tag{4.6.15}$$

where a and b are complex constants. Such 2-component state is called a spinor.

Again, spin is a purely quantum mechanical property without classical analogue. Particles such as the electron, the proton, and the neutron have a half spin. Quarks have a half spin too. Some particles, such as pions, have zero spin. The photon has spin one. Spin is one of the defining intrinsic property of particles. We distinguish two different particles such as the electron from the proton using the difference in their intrinsic properties of mass, charge, and spin. Every electron comes with the same mass, charge, and spin.

Exercise 4.6.2:

Show that the possible values of a measurement of the spin angular momentum along any direction could give only either $+\frac{\hbar}{2}$ or $-\frac{\hbar}{2}$ and nothing else.

Solution: Consider an arbitrary direction with unit vector $\hat{e}_a = a_x\hat{e}_x + a_y\hat{e}_y + a_z\hat{e}_z$. The spin vector operator along this direction is then $\hat{S}_a = \hat{\vec{S}} \cdot \hat{e}_a = a_x\hat{S}_x + a_y\hat{S}_y + a_z\hat{S}_z) = \frac{\hbar}{2}\begin{pmatrix} a_z & a_x - ia_y \\ a_x + ia_y & -a_z \end{pmatrix}$. The eigenvalues are obtained by solving $|\hat{S}_a - \lambda I| = \begin{vmatrix} \frac{\hbar}{2}a_z - \lambda & \frac{\hbar}{2}a_x - i\frac{\hbar}{2}a_y \\ \frac{\hbar}{2}a_x + i\frac{\hbar}{2}a_y & -\frac{\hbar}{2}a_z - \lambda \end{vmatrix} = 0$, which gives the quadratic equation $\lambda^2 - \frac{\hbar^2}{4}a_z^2 - \frac{\hbar^2}{4}a_x^2 - \frac{\hbar^2}{4}a_y^2 = 0$ and the solutions are $\lambda = \pm\frac{\hbar}{2}|\hat{e}_a| = \pm\frac{\hbar}{2}$, since $\hat{e}_a$ is a unit

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vector and $|\hat{e}_a| = 1$. Therefore, the eigenvalues of $\hat{S}_a$ along unit vector $\hat{e}_a$ is $\pm\frac{\hbar}{2}$. Let us write the corresponding state vectors as $|+\rangle$ and $|-\rangle$. To find the vector $|+\rangle$, we solve $\begin{pmatrix} a_z - 1 & a_x - ia_y \\ a_x + ia_y & -a_z - 1 \end{pmatrix}\begin{pmatrix} b \\ c \end{pmatrix} = 0$, which gives $(a_z - 1)b + (a_x - ia_y)c = 0$ and $(a_x + ia_y)b + -(a_z + 1)c = 0$. Using the latter, $\frac{c}{b} = \frac{a_x+ia_y}{a_z+1}$. To write the solutions in simpler form, let us writing the components of $\hat{a}$ using spherical coordinates: $a_x = \sin\theta\cos\phi$, $a_y = \sin\theta\sin\phi$, and $a_z = \cos\theta$. Then, $\frac{c}{b} = \frac{\sin\theta\cos\phi+i\sin\theta\sin\phi}{\cos\theta+1} = \frac{\sin\theta e^{i\phi}}{\cos\theta+1} = \frac{\sin\frac{\theta}{2}e^{i\phi}}{\cos\frac{\theta}{2}}$. Thus, we can choose, $b = \cos\frac{\theta}{2}$ and $c = \sin\frac{\theta}{2}e^{i\phi}$. Repeating similar steps for $|-\rangle$ too, we have

$$|+\rangle = \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2}e^{i\phi} \end{pmatrix}, \qquad |-\rangle = \begin{pmatrix} -\sin\frac{\theta}{2}e^{-i\phi} \\ \cos\frac{\theta}{2} \end{pmatrix} \tag{4.6.16}$$

Exercise 4.6.3:

Determine the matrices for $\hat{\vec{L}}^2$, $\hat{L}_z$, and $\hat{L}_x$ for the four $n = 2$ states of hydrogen in the bases $|nlm_l\rangle = |200\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$, $|211\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$, $|210\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$, and $|21-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$

(5 points)

Exercise 4.6.4: Determine the matrices for $\hat{\vec{J}}^2$, $\hat{J}_z$, $\hat{J}_x$, and $\hat{J}_y$ for $j = \frac{3}{2}$.

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